Classical statistical inference

Uncertainties on arbitrary RV

Part 4

Associated notebook:

04-Basic_statistical_inference_frequentists_2/Frequentist_Monte_Carlo.ipynb

Uncertainty calculation

How to calculate stde (or simply std) on a RV if it is a function of one or multiple RV ?

Case 1: $y = \phi(x)$ $\quad$ $p(x)$ known

Case 2: $z = \phi(x, y)$ $\quad$ $p(x)$ and $p(y)$ known

Uncertainty calculation

Case 1: $y = \phi(x)$ $\qquad$ $p(x)$ known

Example: $y = -2.5 * log(x)$

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$x = \phi^{-1}(y) = 10^{-0.4*y}$

$p(x) \sim U(0,1)$ → 1 if x ∈ [0,1]; 0 if x ∉ [0,1]

$p(y) = 1 \times 0.4\,\ln(10)\,10^{-0.4\,y}$

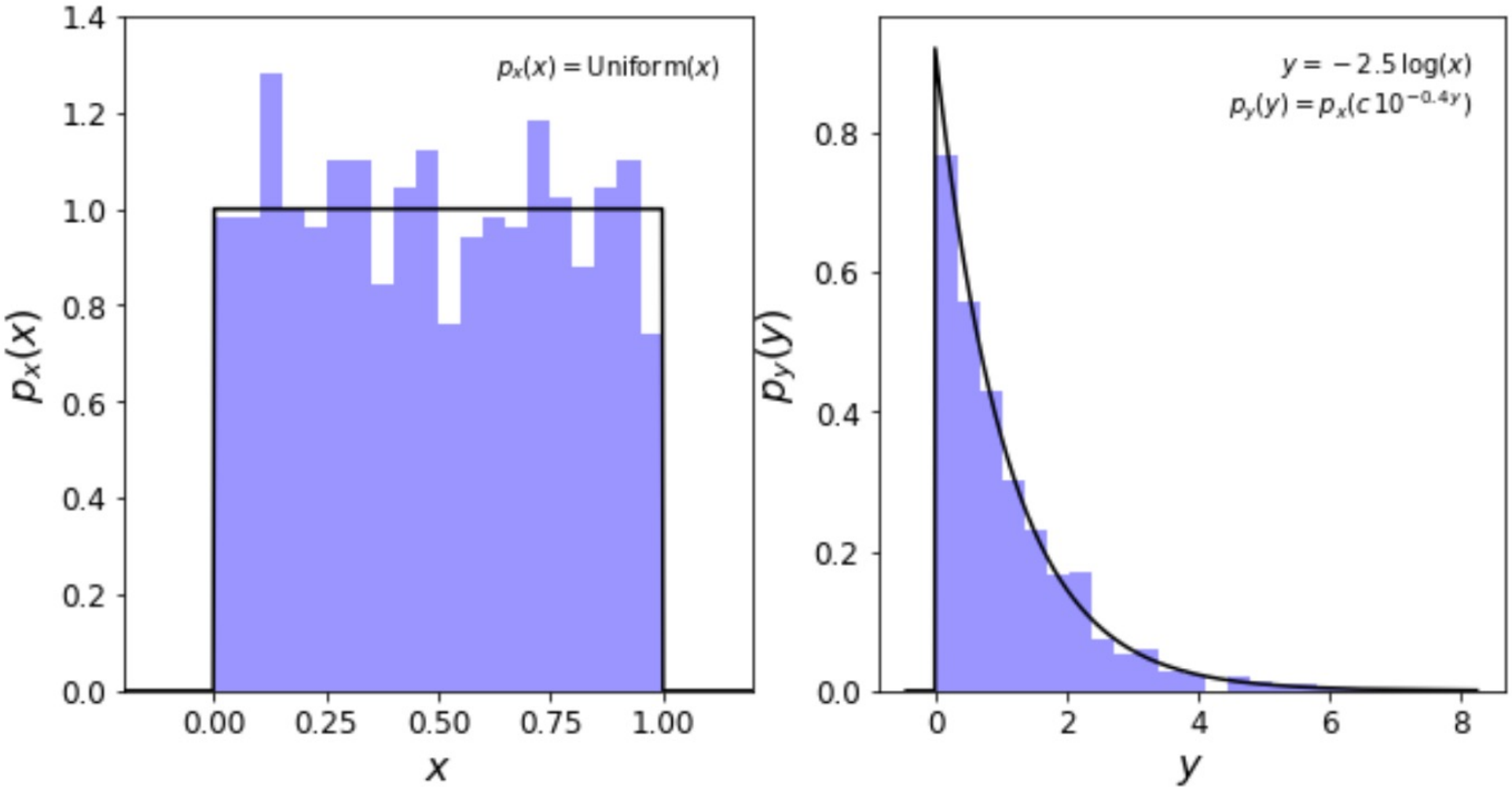
$0 < y < \infty$

Uncertainty calculation

Go to: Sect. II.4.1 of the notebook

See also II.4.1.1 and meaning of dex

Example: $y = -2.5 * log(x)$



Uncertainty calculation

Case 2: $z = \phi(x, y)$ $\qquad$ $p(x)$ and $p(y)$ known

Error propagation formula

$$\sigma_z^2 = \left(\frac{\partial \phi}{\partial x}\right)_{\bar{x}}^2 \sigma_x^2 + \left(\frac{\partial \phi}{\partial y}\right)_{\bar{y}}^2 \sigma_y^2 + 2\left(\frac{\partial \phi}{\partial x}\right)_{\bar{x}} \left(\frac{\partial \phi}{\partial y}\right)_{\bar{y}} \sigma_{xy}$$

Results from Taylor expanding *z* around $\bar{x}$ and $\bar{y}$ => neglects some high order terms

Go to: Sect. II.4.2 of the notebook for the demo

Uncertainty calculation *Monte-Carlo* $c = \Phi(a, b)$: σ_c ?

The expectation ("average") of a function *f(x)* of a RV *x* can be approximated by drawing a virtually infinite sample from *x*

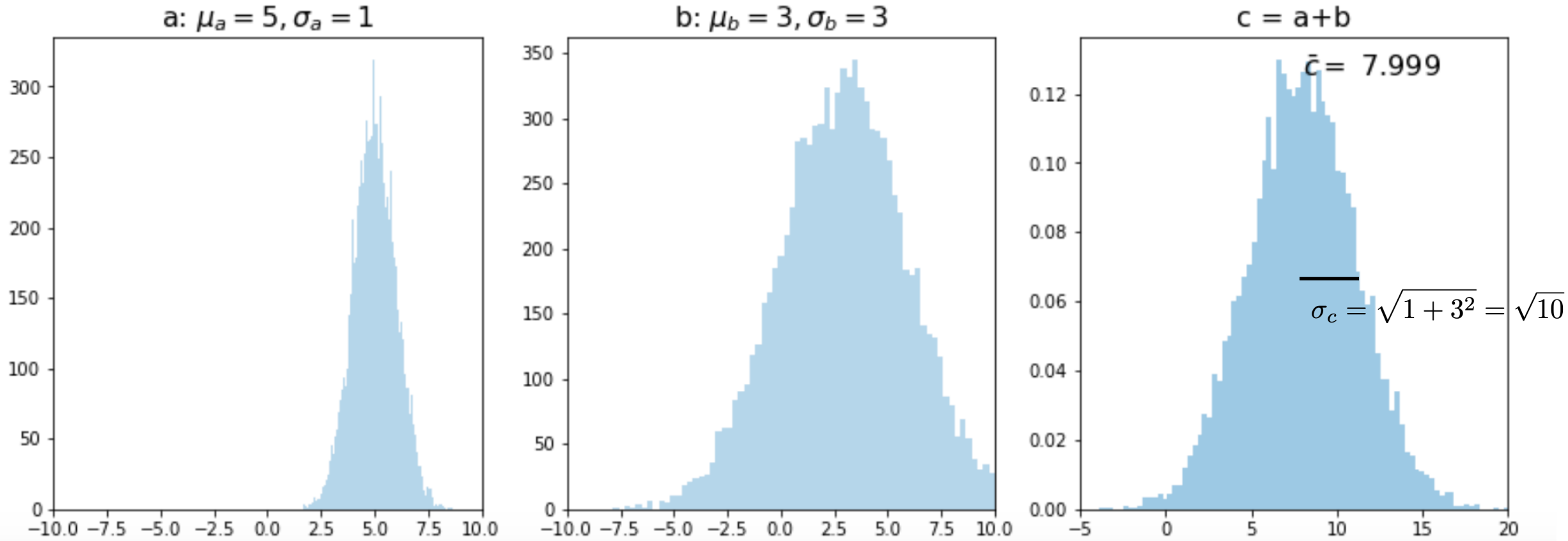
$$< f(x) > \;=\; E(f(x)) = \int_{-\infty}^{+\infty} f(x)h(x)dx \rightarrow \frac{1}{N}\sum_{i}^{N} f(x_i),$$

You can specialise *f() to the calculation of* the mean, or the variance of a RV.

1. Draw random samples from $h(a; \mu_a, \sigma_a)$, $h(b, ; \mu_b, \sigma_b)$
2. Construct a random (*monte-carlo*) sample of $c_i = \Phi(a_i, b_i)$
3. Derive σ_c from your Monte-Carlo sample

Uncertainty calculation *Monte-Carlo*

c = a + b

Go to: Sect. II.4.3 of the notebook

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